Introduction to Econometrics
Week 2 part 1: The algebra of least squares (chapter 1.2)

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Based on Rob Alessie's lecture notes

Outline

The algebra of least squares

- Minimizing the residual sum of squares
- Normal equations
- OLS estimator
- Other concepts (fitted value, residual, R^2, R^2_{un})

The algebra of least squares

- Consider the following population model

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i \tag{1}$$

- Suppose that we have a sample $(\mathbf{x}_i', y_i), i = 1, ..., n$.

  Question: how do we estimate the unknown parameter vector $\boldsymbol{\beta}$ in equation (1)?

  Answer: by means of Ordinary Least Squares

Motivating the name OLS - Simple regression model

- Define **fitted value** for y when $x = x_i$ as

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- Define the **residual** for observation i as the difference between the actual value of y and its fitted value:

$$\hat{u} = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

Note that residuals ARE NOT the same as errors

Motivating the name OLS - Simple regression model

- Let's pick β_0 and β_1 so that the sum of squared residuals is as small as possible:

$$\min_{\beta_0,\beta_1} \sum_i^n \hat{u}_i^2 \Rightarrow \min_{\beta_0,\beta_1} \sum_i^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- The first order conditions are the following:

$$\begin{aligned} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= \frac{\sum_i x_i (y_i - \bar{y})}{\sum_i x_i (x_i - \bar{x})} \end{aligned}$$

Multivariate case: the algebra of least squares

- First consider the following arbitrary linear combination:

$$\tilde{\beta}_1 x_{i1} + \tilde{\beta}_2 x_{i2} + \ldots + \tilde{\beta}_K x_{iK} = \mathbf{x}_i' \tilde{\boldsymbol{\beta}} \tag{2}$$

  where $\tilde{\boldsymbol{\beta}} = (\tilde{\beta}_1, \ldots, \tilde{\beta}_K)'$ is to be chosen.
- Consider the difference between y_i and the *linear approximation* $\mathbf{x}_i' \tilde{\boldsymbol{\beta}}$

$$y_i - \mathbf{x}_i' \tilde{\boldsymbol{\beta}} \tag{3}$$

- From these differences, form the **sum of squared residuals (SSR)**

$$SSR(\tilde{\boldsymbol{\beta}}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i' \tilde{\boldsymbol{\beta}})^2 = (\mathbf{y} - \mathbf{X}\tilde{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\tilde{\boldsymbol{\beta}}) \tag{4}$$

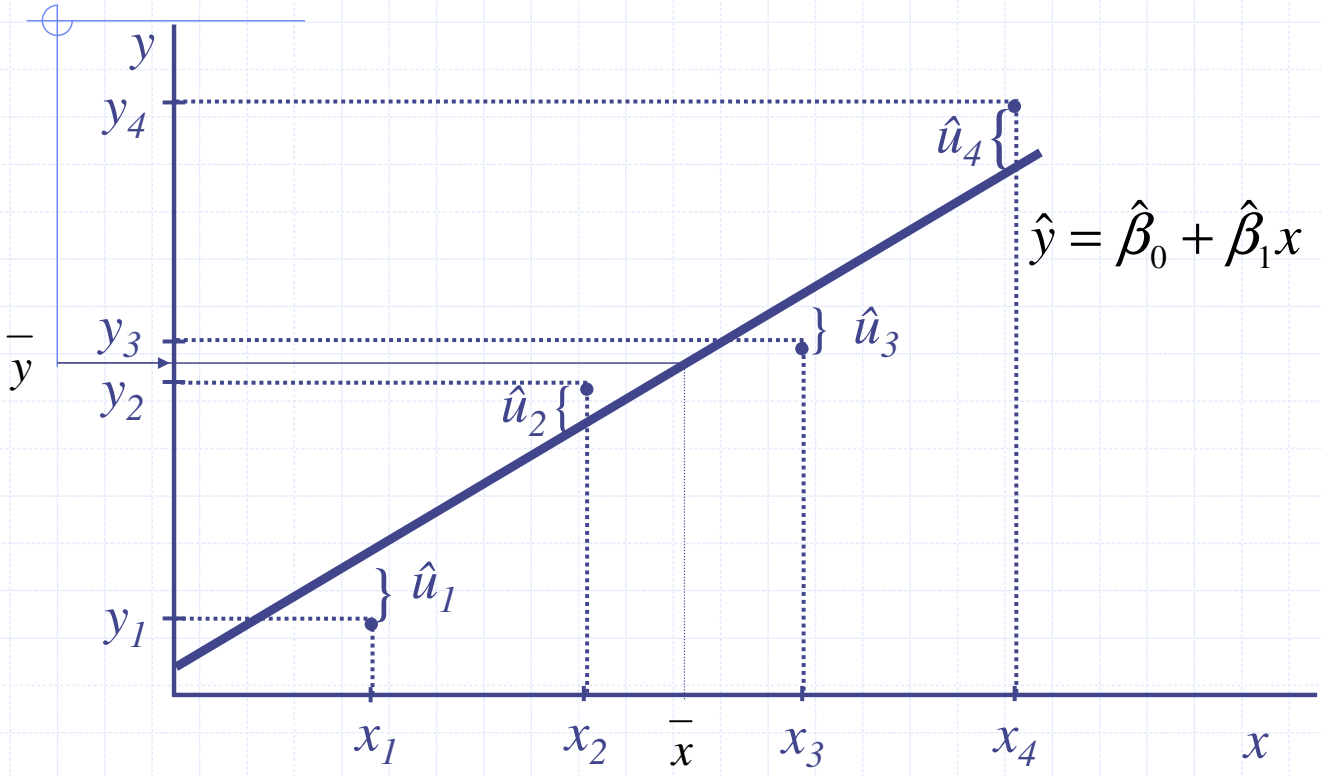
  This sum is also called the **residual sum of squares (RSS)**

The algebra of least squares

- **Approach of Ordinary Least Squares (OLS)**: The OLS estimate, **b**, of $\boldsymbol{\beta}$ is the $\tilde{\boldsymbol{\beta}}$ that minimizes function (4) (see the graphical illustration):

$$\mathbf{b} \equiv \underset{\tilde{\boldsymbol{\beta}}}{\operatorname{argmin}}\, SSR(\tilde{\boldsymbol{\beta}}) \tag{5}$$

Sample regression line (K=2), sample data points and the associated residuals



Normal equations

- To solve minimization problem (5), is to derive the first order conditions (F.O.C.), by setting the partial derivatives equal to zero.
- Rewrite (4) as follows:

$$
\begin{aligned}
SSR(\tilde{\beta}) &= (\mathbf{y} - \mathbf{X}\tilde{\beta})'(\mathbf{y} - \mathbf{X}\tilde{\beta}) \\
&= (\mathbf{y}' - \tilde{\beta}'\mathbf{X}')(\mathbf{y} - \mathbf{X}\tilde{\beta}) \\
&= \mathbf{y}'\mathbf{y} - \tilde{\beta}'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\tilde{\beta} + \tilde{\beta}'\mathbf{X}'\mathbf{X}\tilde{\beta} \\
&= \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\tilde{\beta} + \tilde{\beta}'\mathbf{X}'\mathbf{X}\tilde{\beta} \\
&\equiv \mathbf{y}'\mathbf{y} - 2\mathbf{a}'\tilde{\beta} + \tilde{\beta}'\mathbf{A}\tilde{\beta}
\end{aligned} \tag{6}
$$

where $\mathbf{a} = \mathbf{X}'\mathbf{y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$.

Normal equations

Recalling from matrix algebra (cf. handout Matrix Algebra) that

$$\frac{\partial(\mathbf{a}'\tilde{\boldsymbol{\beta}})}{\partial\tilde{\boldsymbol{\beta}}} = \mathbf{a} \text{ and } \frac{\partial(\tilde{\boldsymbol{\beta}}'\mathbf{A}\tilde{\boldsymbol{\beta}})}{\partial\tilde{\boldsymbol{\beta}}} = 2\mathbf{A}\tilde{\boldsymbol{\beta}} \text{ for } \mathbf{A} \text{ symmetric}$$

the K dimensional vector of partial derivatives becomes

$$\frac{\partial SSR(\tilde{\boldsymbol{\beta}})}{\partial\tilde{\boldsymbol{\beta}}} = -2\mathbf{a} + 2\mathbf{A}\tilde{\boldsymbol{\beta}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\tilde{\boldsymbol{\beta}} \tag{7}$$

Normal equations

- The first order conditions:

$$-2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\mathbf{b} = 0 \tag{8}$$

  or

$$(\mathbf{X}'\mathbf{X})\mathbf{b} = \mathbf{X}'\mathbf{y} \tag{9}$$

  Here, we replaced $\tilde{\beta}$ by $\mathbf{b}$ because the OLS estimate $\mathbf{b}$ is the $\tilde{\beta}$ that satisfies the first order conditions.
  The K equations (9) are called the **normal equations**

- The first order conditions (9) are just necessary. The second order condition states that the Hessian

$$\frac{\partial^2 SSR(\tilde{\beta})}{\partial\tilde{\beta}\partial\tilde{\beta}'} = 2\mathbf{X}'\mathbf{X} \tag{10}$$

  should be positive definite at the optimum.

- Indeed $\mathbf{X}'\mathbf{X}$ is a positive definite matrix so that $\mathbf{b}$ is the unique solution of the minimization problem (4)

Normal equations

The normal equations can also be written as follows (check!)

$$\left(\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i'\right) \mathbf{b} = \sum_{i=1}^{n} \mathbf{x}_i y_i \tag{11}$$

- Note that the system of normal equations linear in $\mathbf{b}$
- System of equations (9) (FOC's) provides a unique solution as long as the **no perfect multicollinearity assumption** 1.3 holds, i.e. the following matrix is of full column rank: $\mathbf{X}'\mathbf{X} = (\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i')$.
- In that case, the solution of this problem, denoted by $\mathbf{b}$ is given by:

$$\mathbf{b} = \left(\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i'\right)^{-1} \sum_{i=1}^{n} \mathbf{x}_i y_i = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \tag{12}$$

More concepts

- **Best linear approximation (fitted value)**

$$\hat{y}_i = \mathbf{x}_i'\mathbf{b} \tag{13}$$

- **Residuals**
  $(n \times 1)$ vector of **(OLS) residuals**

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\mathbf{b} \tag{14}$$

  Its i-th element $e_i = y_i - \mathbf{x}_i'\mathbf{b}$
- **Residual sum of squares** (=minimum value objective function $SSR(\tilde{\beta})$ (cf equation (4)):

$$SSR = SSR(\mathbf{b}) = \mathbf{e}'\mathbf{e} = \sum_{i=1}^{n} e_i^2 \tag{15}$$

More concepts

Notice that we can rewrite first order conditions (9) as follows:

$$\mathbf{X}'(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0} \text{ or } \mathbf{X}'\mathbf{e} = \mathbf{0} \text{ or} \tag{16a}$$

$$\frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i e_i = \mathbf{0} \text{ or } \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i (y_i - \mathbf{x}_i'\mathbf{b}) = \mathbf{0} \tag{16b}$$

If $\mathbf{x}_i$ contains a constant then (16a) implies that

$$\sum_{i=1}^{n} e_i = 0$$

and that

$$\overline{y} = \overline{\mathbf{x}}'\mathbf{b}$$

More concepts

- We already know that $E(\varepsilon_i|\mathbf{X}) = 0$ (strict exogeneity) implies the following K **moment conditions**:

$$E(\mathbf{x}_i\varepsilon_i) = 0$$

  (see notes previous lecture)
- Equation (16b) can be viewed as the sample analogue of these moment conditions.
- One can therefore view OLS as a **method of moment estimator** (see chapter 3 of the book)
- MM estimation chooses as the estimate of the unknown parameter vector θ the value of θ that causes estimated moments in the data to equal their sample counterparts

More concepts

- $(n \times n)$ **Projection matrix P**

$$\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \tag{17}$$

- **Annihilator matrix**

$$\mathbf{M} = \mathbf{I}_n - \mathbf{P} \tag{18}$$

- Properties of the projection and annihilator matrices **P** and **M**
  - Both **P** and **M** are symmetric and idempotent.
  - $\mathbf{PX} = \mathbf{X}$
  - $\mathbf{MX} = \mathbf{0}$
  - $\mathbf{Py} = \hat{\mathbf{y}}$
  - $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{P})\mathbf{y} = \mathbf{My} = \mathbf{M}(\mathbf{X}\beta + \varepsilon) = \mathbf{M}\varepsilon$
  - $SSR = \mathbf{e}'\mathbf{e} = \varepsilon'\mathbf{M}'\mathbf{M}\varepsilon = \varepsilon'\mathbf{M}\varepsilon$

You should check the validity of these statements (see handout Matrix Algebra)

More concepts

We know how to estimate the unknown vector of the coefficients, but we still do not know how to get an estimate of the variance of the error term (σ^2)

s^2: OLS estimate of σ^2 (the variance of the error term)

$$s^2 = \frac{SSR}{n-K} = \frac{\mathbf{e}'\mathbf{e}}{n-K} \tag{19}$$

standard error of the regression (SER): $s = \sqrt{s^2}$.
It is an estimate of the standard error σ of the error term.

More concepts

Sampling error: $\mathbf{b} - \boldsymbol{\beta}$.
It can be related to ε (the error term) as follows:

$$\begin{aligned}\mathbf{b} - \boldsymbol{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} - \boldsymbol{\beta} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) - \boldsymbol{\beta} \\ &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} - \boldsymbol{\beta} \\ &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} - \boldsymbol{\beta} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\end{aligned} \tag{20}$$

Goodness of fit measures

Uncentered R^2

Variability of the dependent variable can be measured by $\sum y_i^2 = \mathbf{y}'\mathbf{y}$.
We can decompose $\mathbf{y}'\mathbf{y}$ as follows:

$$\begin{aligned}\mathbf{y}'\mathbf{y} &= (\hat{\mathbf{y}} + \mathbf{e})'(\hat{\mathbf{y}} + \mathbf{e}) \\ &= \hat{\mathbf{y}}'\hat{\mathbf{y}} + 2\hat{\mathbf{y}}'\mathbf{e} + \mathbf{e}'\mathbf{e} \\ &= \hat{\mathbf{y}}'\hat{\mathbf{y}} + 2\mathbf{b}'\mathbf{X}'\mathbf{e} + \mathbf{e}'\mathbf{e} \\ &= \hat{\mathbf{y}}'\hat{\mathbf{y}} + \mathbf{e}'\mathbf{e}\end{aligned} \tag{21}$$

In the last step I use the property that $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (by the normal equation)

Goodness of fit measures

Definition **uncentered** R^2 :

$$R^2_{uc} = 1 - \frac{\mathbf{e}'\mathbf{e}}{\mathbf{y}'\mathbf{y}} \tag{22}$$

Because of the decomposition (21) this equals

$$R^2_{uc} = \frac{\hat{\mathbf{y}}'\hat{\mathbf{y}}}{\mathbf{y}'\mathbf{y}} \tag{23}$$

Since both $\hat{\mathbf{y}}'\hat{\mathbf{y}}$ and $\mathbf{y}'\mathbf{y}$ are nonnegative, $0 \leq R^2_{uc} \leq 1$
Thus, the uncentered R^2 has the interpretation of the fraction of the variation of the dependent variable that is attributable to the variation in the explanatory variables.

Goodness of fit measures

Centered R^2 (coefficient of determination)

- If the only regressor is a constant, the $b = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ (sample average of the dependent variable). In that case
  - $\hat{y}_i = \bar{y}$
  - $\hat{\mathbf{y}}'\hat{\mathbf{y}} = n\bar{y}^2$
  - $\mathbf{e}'\mathbf{e} = \sum_{i=1}^{n}(y_i - \bar{y})^2$
- If the regressors also include nonconstant variables, it can be shown that $\sum_i (y_i - \bar{y})^2$ can be decomposed as follows (check!):

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n} e_i^2 \tag{24}$$

Goodness of fit measures

- The **coefficient of determination**, R^2, is defined as

$$R^2 \equiv 1 - \frac{\sum_{i=1}^{n} e_i^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{25}$$

Because of the decomposition (24) this equals

$$R^2 = \frac{\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{26}$$

- It holds that $0 \leq R^2 \leq 1$

Goodness of fit measures

- Interpretation R^2: $100*R^2$ is percentage of sample variance in y explained by the nonconstant regressors (model).
  - low R^2: bad fit of the model
  - high R^2: good fit of the model
- Be careful with interpretation of the R^2!!! Value of R^2 increases by the inclusion of extra irrelevant regressors.
- **Adjusted** R^2**,** $\bar{R}^2$, addresses this problem somewhat

$$\bar{R}^2 = 1 - \frac{\mathbf{e}'\mathbf{e}/(n-K)}{(\sum_{i=1}^{n}(y_i - \bar{y})^2)/(n-1)} \tag{27}$$